

⌘ · MCMC and Variational Inference

1 You have the posterior. Now what?

The previous chapter ended with Bayes' theorem:

$$p(\theta | x) = \frac{p(x | \theta) p(\theta)}{p(x)} \quad (1.1)$$

The posterior $p(\theta | x)$ is the answer to inference. But the posterior is a *distribution*, not a number. For two parameters, it's a surface. For fifteen parameters, it's a 15-dimensional landscape.

Summaries are integrals. We rarely need the full distribution. We want summaries: the posterior mean $\mathbb{E}[\theta | x]$, the variance, credible intervals. Each of these is an integral:

$$\mathbb{E}[\theta | x] = \int \theta p(\theta | x) d\theta \quad (1.2)$$

This says: average over all possible θ values, weighting each by how probable it is given the data.

The problem: we can't do these integrals. For simple models (like the Gaussian example in Chapter 1), these integrals have closed-form solutions. But for almost any interesting model, they don't. We can *evaluate* the posterior at any point, but we cannot *integrate* it analytically.

This is the computational bottleneck of Bayesian inference: we have a formula for the posterior, but we're stuck behind integrals we cannot solve.

Two strategies.

1. **MCMC (Markov chain Monte Carlo):** Generate samples from the posterior and replace integrals with averages: $\mathbb{E}[\theta | x] \approx \frac{1}{N} \sum_i \theta_i$. Exact in the limit of infinite samples, but can be slow.
2. **Variational inference:** Approximate the posterior with a simpler distribution $q(\theta) \approx p(\theta | x)$ that we *can* integrate. Fast, but only approximate.

2 Monte Carlo integration

If we had samples from the posterior, we could replace integrals with averages. Suppose we obtain $\theta_1, \theta_2, \dots, \theta_N \sim p(\theta | x)$. Then:

$$\mathbb{E}[f(\theta) | x] = \int f(\theta) p(\theta | x) d\theta \approx \frac{1}{N} \sum_{i=1}^N f(\theta_i) \quad (2.1)$$

Want the posterior mean? Set $f(\theta) = \theta$ and average your samples. Want the variance? Compute the sample variance. Want a credible interval? Sort the samples and find the 2.5th and 97.5th percentiles.

Sampling converts integration into averaging.

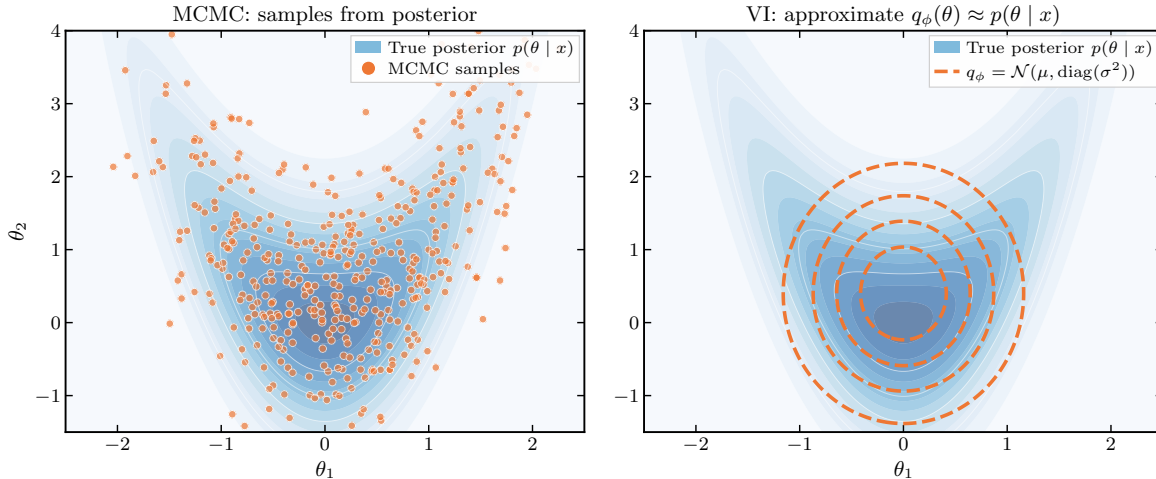


Figure 1. Two approaches to intractable posteriors. The blue shading shows a curved “banana” posterior $p(\theta | x)$. (Left) MCMC generates samples distributed according to the true posterior, capturing its full structure including correlations. (Right) Variational inference approximates the posterior with a simpler distribution—here a mean-field Gaussian $q_\phi(\theta) = \mathcal{N}(\mu, \text{diag}(\sigma^2))$. The approximation finds the right location but cannot represent the curvature.

Why does averaging work? The law of large numbers. Each $f(\theta_i)$ is a draw from some distribution with mean $\mu = \mathbb{E}[f(\theta)]$ —the integral we want. The sample average converges to μ as $N \rightarrow \infty$.

How fast? The standard error is $\sigma/\sqrt{N}$, where $\sigma^2 = \text{Var}[f(\theta)]$. To halve the error, quadruple the samples.

Dimension-free convergence. The error $\sigma/\sqrt{N}$ depends on N and σ , but *not on the dimension of θ* . Compare this to grid integration: with 10 points per dimension, a 15D integral needs 10^{15} evaluations. Grid methods scale exponentially with dimension; Monte Carlo doesn’t.

There’s a catch: the dimension affects how hard it is to *get* the samples. But once we have them, the averaging is dimension-free.

The remaining question. How do we sample from a distribution we can only evaluate up to a normalizing constant? We can compute $p(x | \theta)p(\theta)$ for any θ , but we don’t know $p(x)$, so we can’t normalize.

This is where MCMC comes in.

3 The Metropolis algorithm

Imagine exploring a landscape where elevation represents probability—higher means more probable. You want to collect samples that represent this landscape (spending more time in high-probability regions), but you can only check elevation where you’re standing.

Historical note: Arianna Rosenbluth.— The algorithm first appeared in a 1953 paper by Metropolis, the Rosenbluths, and the Tellers. Marshall Rosenbluth later acknowledged that “Arianna did all the coding”—programming in raw machine language on the MANIAC I computer at Los Alamos. She was 24 years old with a PhD in physics from Harvard (only the fifth woman to do so). More than 50,000 papers have since cited this work.

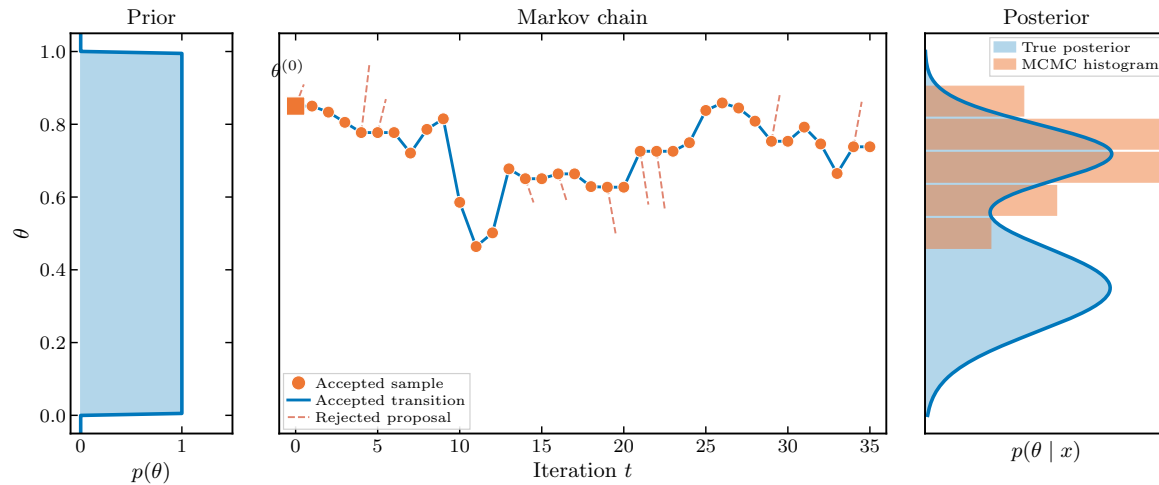


Figure 2. The Metropolis algorithm in action. Starting from a uniform prior $p(\theta)$ (left), the algorithm builds a Markov chain (center) by proposing moves and accepting or rejecting them based on the posterior ratio. Accepted proposals (blue lines) move the chain to a new state; rejected proposals (red dashed) leave it in place. After many iterations, the histogram of visited states (orange bars, right) approximates the true posterior $p(\theta | x)$ (blue curve).

The Metropolis algorithm is a random walk with a clever acceptance rule:

1. **Propose:** From current position θ , propose a move to $\theta' \sim \mathcal{N}(\theta, \sigma^2 I)$ —a Gaussian centered at the current position.
2. **Accept or reject:** Compute the acceptance probability:

$$\alpha = \min \left(1, \frac{p(\theta' | x)}{p(\theta | x)} \right) \quad (3.1)$$

If the new spot has higher probability: always accept. If lower: accept with probability α .

3. **Record:** If accepted, move to θ' ; otherwise stay at θ . Record the current position as a sample.

Accepting “downhill” moves with probability proportional to the density ratio prevents getting stuck on one peak while still spending more time in high-probability regions.

3.1 Why does this work?

Why this particular acceptance rule? The short answer: accepting with probability $\alpha = \min(1, p(\theta')/p(\theta))$ ensures the chain spends time in each region proportional to its probability.

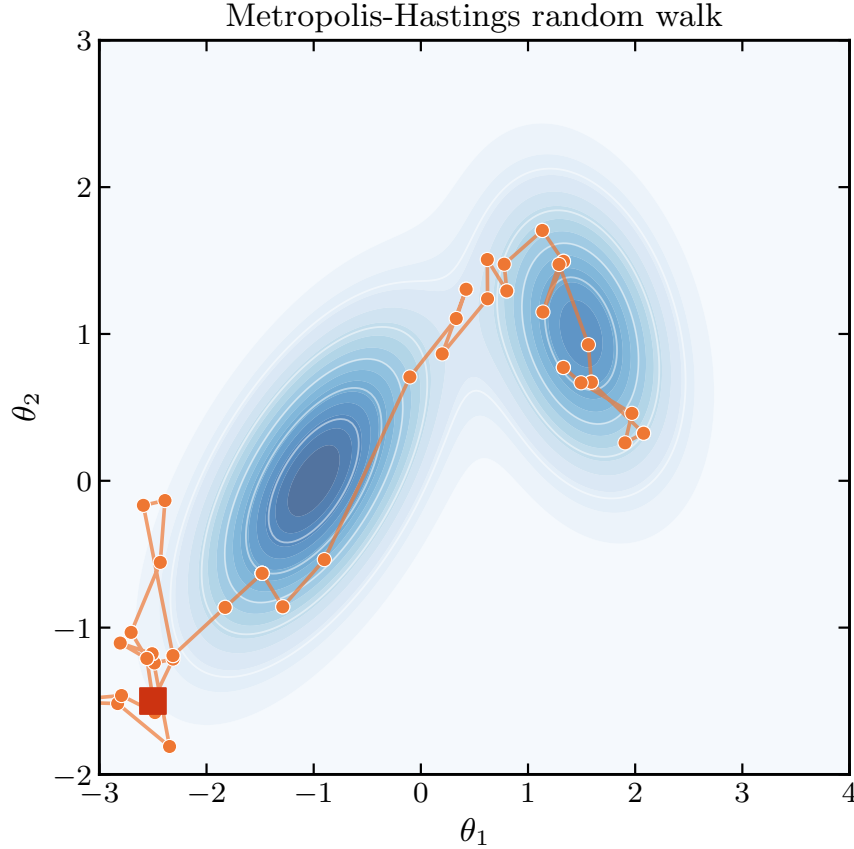


Figure 3. MCMC as landscape exploration. The surface represents a posterior distribution over two parameters. The algorithm proposes moves (arrows), accepts those that go uphill or pass a random test when going downhill, and traces out a path that spends more time in high-probability regions.

Normalizing constants cancel. The acceptance ratio only involves $p(\theta' | x)/p(\theta | x)$. Since both share the same normalizing constant $p(x)$, it cancels:

$$\frac{p(\theta' | x)}{p(\theta | x)} = \frac{p(x | \theta') p(\theta')}{p(x | \theta) p(\theta)} \quad (3.2)$$

We only need the unnormalized posterior—likelihood times prior.

Detailed balance (optional).— The formal justification is *detailed balance*: in equilibrium, the flow of probability between any two states must balance. For states θ_A and θ_B :

$$p(\theta_A) T(\theta_A \rightarrow \theta_B) = p(\theta_B) T(\theta_B \rightarrow \theta_A) \quad (3.3)$$

where T is the transition probability (propose times accept). The Metropolis acceptance ratio makes this hold. Suppose $p(\theta') \geq p(\theta)$. We always accept $\theta \rightarrow \theta'$, but only accept $\theta' \rightarrow \theta$ with probability $p(\theta)/p(\theta')$. The flows balance:

$$p(\theta) \cdot 1 = p(\theta') \cdot \frac{p(\theta)}{p(\theta')} = p(\theta) \quad \checkmark \quad (3.4)$$

4 Tuning and convergence

The proposal distribution has a free parameter: step size σ . The choice matters.

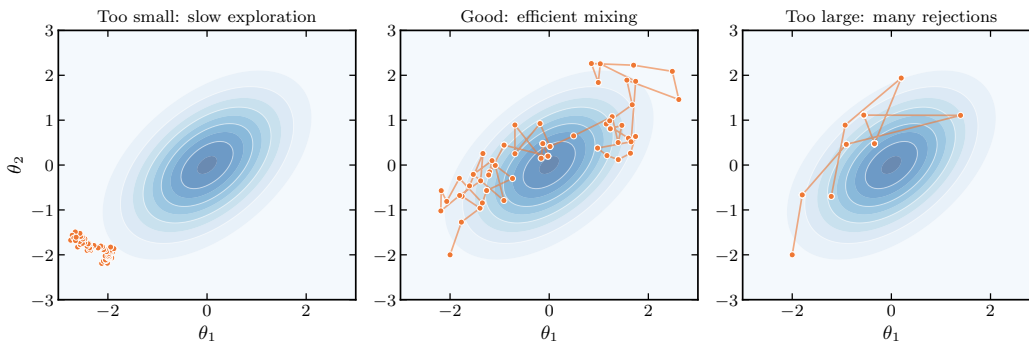


Figure 4. The step size tradeoff. (a) Step size too small: nearly all proposals are accepted, but exploration is slow. (b) Step size too large: most proposals land in low-probability regions and are rejected. (c) Step size well-tuned: efficient exploration.

- **Too small:** Nearly all proposals are accepted, but exploration is slow.
- **Too large:** Most proposals land in low-probability regions and are rejected.

A good acceptance rate is roughly 25% in high dimensions. The intuition: at this rate, the product (acceptance rate) $\times$ (step size) is maximized.

Autocorrelation. MCMC samples are correlated—each is near the previous. The **autocorrelation time** τ measures how many steps until samples become approximately independent. If $\tau = 100$, then every 100th sample is effectively independent; the effective sample size is $N_{\text{eff}} = N/\tau$.

5 The failure of random walks in high dimensions

The random walk Metropolis algorithm scales poorly with dimension. The autocorrelation time grows as $\tau \propto d$ —in d dimensions, you need d times more samples.

Why? In high dimensions, probability concentrates in a thin shell called the *typical set*. For a d -dimensional Gaussian, this shell lies at radius $r \sim \sqrt{d}$ from the mode, with thickness scaling as $1/\sqrt{d}$.

A random walker on this shell faces a geometric problem. To explore the posterior, we need to traverse the shell—move along its surface. But a random walk proposes moves in all directions equally. Most directions point off the shell into low-probability regions.

To stay on the thin shell, we need small step sizes: $\sigma \propto 1/\sqrt{d}$. But then the tangential component—the part that actually explores—also shrinks. The chain makes tiny steps, and autocorrelation time grows as $\tau \propto d$.

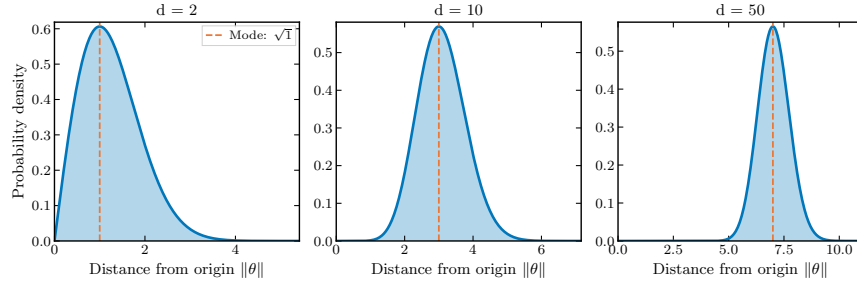


Figure 5. The typical set in high dimensions. For a d -dimensional Gaussian, most probability mass lies not at the mode (where density is highest) but in a thin shell at radius $\sqrt{d}$. Volume grows rapidly with radius, overwhelming the density decay.

6 Hamiltonian Monte Carlo

The posterior has structure—the gradient $\nabla \log p(\theta | x)$ tells us which way is “uphill.” The Metropolis random walk ignores this information. Hamiltonian Monte Carlo (HMC) uses the gradient to make informed, directed moves.

6.1 The key idea

The random walk’s problem is that it doesn’t know which directions stay on the typical set versus which directions point off into low-probability regions. The gradient tells us: $\nabla \log p(\theta)$ points toward higher probability (radially inward), so directions perpendicular to the gradient are tangent to level sets—exactly where we want to move.

But we can’t just move perpendicular to the gradient, because that direction changes as we move. We need to *follow* the contours of the distribution as they curve. Physics gives us this: Hamiltonian dynamics.

6.2 The physics picture

Imagine a ball rolling on a surface where height is negative log-probability (valleys are high-probability regions). Give the ball a push, and it rolls: speeding up downhill, slowing uphill, curving along the contours. A ball rolling without friction conserves energy—it stays on a single energy level, never drifting into low-probability regions.

Hamiltonian dynamics naturally traverses level sets of probability.

Augment parameters θ with momentum ρ . The Hamiltonian (total energy) is:

$$H(\theta, \rho) = \underbrace{-\log p(\theta | x)}_{\text{potential energy}} + \underbrace{\frac{1}{2}\rho^\top \rho}_{\text{kinetic energy}} \quad (6.1)$$

Hamilton’s equations give the dynamics:

$$\frac{d\theta}{dt} = \rho \quad (\text{position changes according to momentum}) \quad (6.2)$$

$$\frac{d\rho}{dt} = \nabla \log p(\theta | x) \quad (\text{momentum changes according to gradient}) \quad (6.3)$$

6.3 The algorithm

1. **Sample momentum:** Draw $\rho \sim \mathcal{N}(0, I)$ —give the ball a random push.
2. **Simulate dynamics:** Integrate Hamilton’s equations for L steps of size ϵ , using the *leapfrog* integrator:

$$\rho_{t+\epsilon/2} = \rho_t + \frac{\epsilon}{2} \nabla \log p(\theta_t \mid x) \quad (6.4)$$

$$\theta_{t+\epsilon} = \theta_t + \epsilon \rho_{t+\epsilon/2} \quad (6.5)$$

$$\rho_{t+\epsilon} = \rho_{t+\epsilon/2} + \frac{\epsilon}{2} \nabla \log p(\theta_{t+\epsilon} \mid x) \quad (6.6)$$

Leapfrog is *symplectic*: it nearly conserves energy, keeping the acceptance rate high.

3. **Metropolis correction:** Accept with probability $\min(1, e^{-H(\theta', \rho') + H(\theta, \rho)})$. If the integrator were exact, we’d always accept. The Metropolis step corrects for discretization errors.

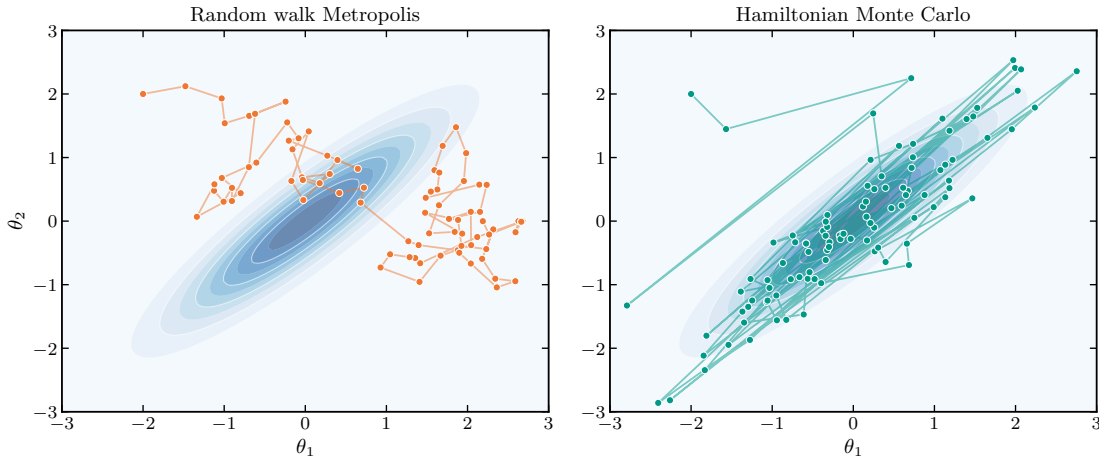


Figure 6. Metropolis random walk vs. Hamiltonian Monte Carlo. (a) Metropolis: the chain diffuses slowly, taking many small steps. (b) HMC: gradient-informed trajectories make long, directed moves across the posterior.

HMC achieves autocorrelation time τ nearly independent of dimension—a large improvement over random walk ($\tau \propto d$). The cost is requiring gradients, but automatic differentiation makes this routine.

7 When to use what

The fundamental tradeoff. Every MCMC algorithm faces the same tension: *propose far* to explore quickly, but *propose carefully* to avoid rejection.

When gradients are available. If you can compute $\nabla \log p(\theta)$ —and with automatic differentiation, you usually can—use HMC or NUTS (which auto-tunes trajectory length).

When gradients are not available. If the likelihood is a black box (simulator, legacy code), gradient-free methods are required: random walk Metropolis, affine-invariant ensemble samplers (emcee), or methods that learn a surrogate model.

8 Diagnostics

We can never prove MCMC has converged, but failures can often be detected.

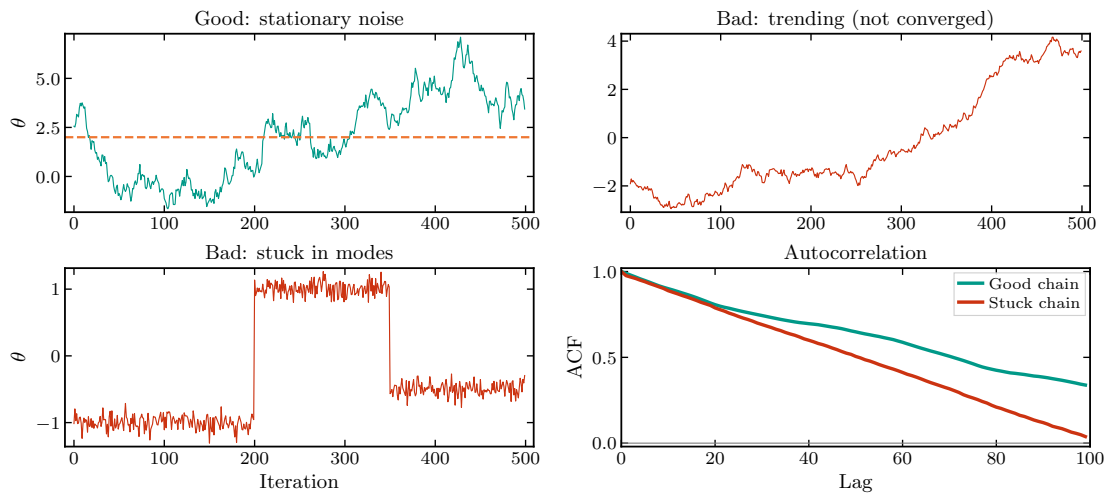


Figure 7. MCMC diagnostics. (a) Good mixing: the chain fluctuates around a stable mean with no trends. (b) Poor mixing: slow drift or getting stuck indicates the sampler needs tuning.

Trace plots. Plot each parameter against iteration number. A converged chain looks like stationary noise.

Multiple chains. Run several chains from different starting points. The $\hat{R}$ statistic compares within-chain variance to between-chain variance; $\hat{R} \approx 1$ indicates agreement.

Effective sample size. $N_{\text{eff}} = N/\tau$: from N correlated samples, we have information equivalent to N_{eff} independent ones. Report this, not the raw sample count.

9 Using samples

Once we have posterior samples, we can answer any question about the posterior.

Point estimates. Posterior mean: $\bar{\theta} = \frac{1}{N} \sum_i \theta_i$.

Credible intervals. The 16th and 84th percentiles give 68% credible intervals; 2.5th and 97.5th give 95%.

Marginalization is trivial. To get $p(\theta_1 | x)$, just histogram the θ_1 values from your samples. What would be a nightmare integral becomes a one-liner with samples.

Derived quantities. Want the posterior over some function $g(\theta)$? Evaluate g on each sample and histogram the results.

Work in log-probability.— Likelihoods are products of many terms. In any real dataset, this product underflows to zero. Always work with log-probabilities:

$$\log p(x \mid \theta) = \sum_i \log p(x_i \mid \theta) \quad (9.1)$$

The acceptance ratio becomes a difference: $\log \alpha = \log p(\theta') - \log p(\theta)$. This is mandatory for any practical implementation.

10 Visualizing high-dimensional posteriors

In real problems, the posterior lives in many dimensions. Cosmological parameter estimation involves 6+ parameters; gravitational wave inference involves 15+.

Corner plots. The standard visualization is the *corner plot*: a grid showing all pairwise 2D marginals (off-diagonal) and 1D marginals (diagonal).

Corner plots reveal structure that summary statistics hide:

- **Degeneracies:** Parameters that trade off against each other appear as tilted ellipses or curved “bananas.”
- **Multimodality:** Multiple peaks indicate distinct solutions.
- **Non-Gaussianity:** Asymmetric marginals suggest the posterior is far from Gaussian.

11 Variational inference

MCMC gives exact samples but can be slow—especially for models with many parameters or applications requiring real-time inference.

Variational inference (VI) trades exactness for speed. Instead of sampling, we approximate the posterior with a simpler distribution. The result is an optimization problem.

The idea. Choose a family $\mathcal{Q}$ of tractable distributions (e.g., Gaussians) and find the member q^* that best approximates the true posterior.

Measuring closeness. The Kullback–Leibler divergence measures how much information is lost when we use q to approximate p :

$$\text{KL}(q \parallel p) = \int q(\theta) \log \frac{q(\theta)}{p(\theta \mid x)} d\theta \quad (11.1)$$

We want $q^* = \arg \min_{q \in \mathcal{Q}} \text{KL}(q(\theta) \parallel p(\theta \mid x))$. But computing the KL requires the normalizing constant $p(x)$ —exactly the intractable integral we’re trying to avoid.

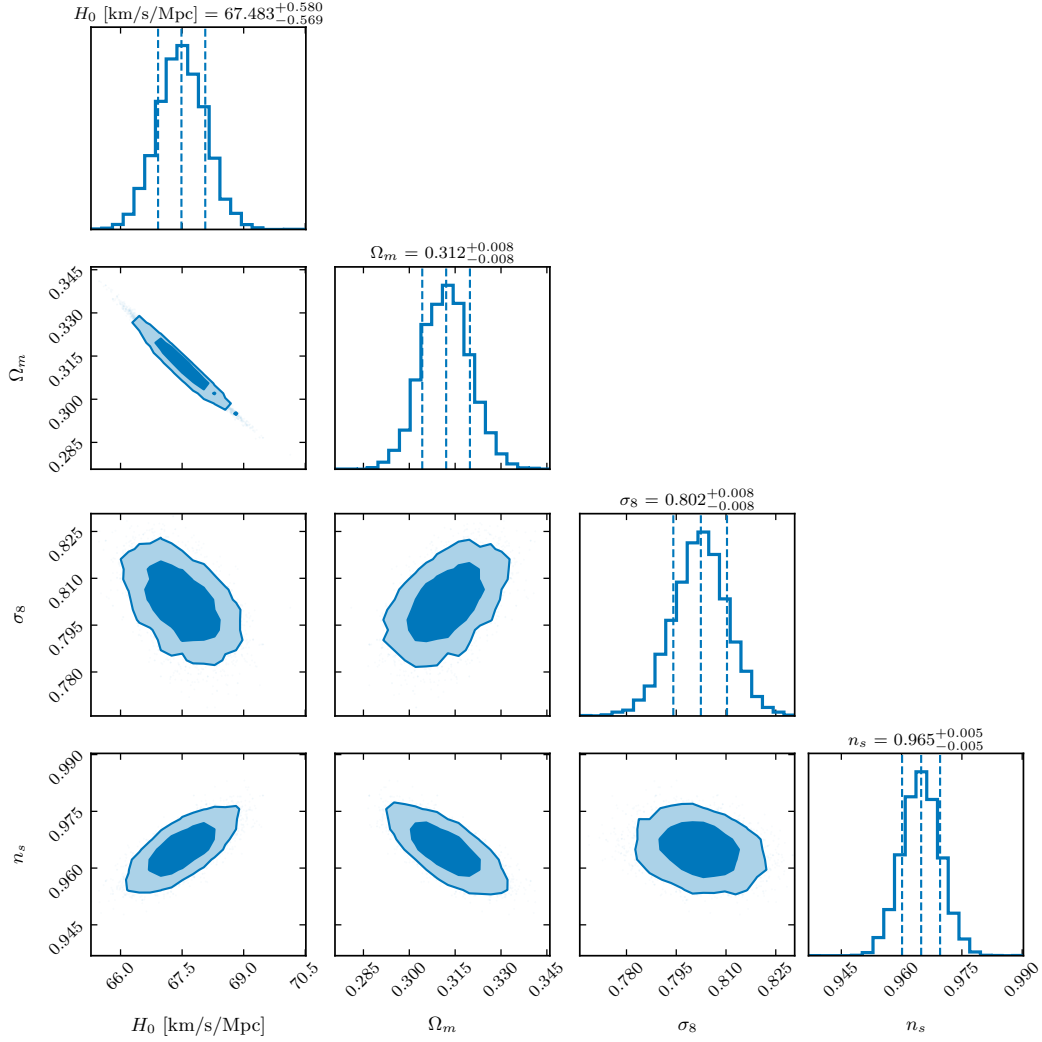


Figure 8. Corner plot from Planck CMB parameter estimation. Parameters shown: Hubble constant H_0 , matter density Ω_m , fluctuation amplitude σ_8 , and spectral index n_s . The H_0 - Ω_m anticorrelation reflects how both affect the angular diameter distance to the CMB. Contours show 68% and 95% credible regions.

12 The evidence lower bound (ELBO)

The key identity is:

$$\log p(x) = \text{ELBO}(q) + \text{KL}(q||p) \quad (12.1)$$

where $\text{ELBO}(q) = \mathbb{E}_q[\log p(x, \theta)] - \mathbb{E}_q[\log q(\theta)]$.

Since $\text{KL} \geq 0$, the ELBO is a lower bound on the log-evidence. **Maximizing the ELBO is equivalent to minimizing the KL divergence**, since $\log p(x)$ is fixed. And the ELBO only involves the joint $p(x, \theta) = p(x | \theta)p(\theta)$ —no normalizing constant needed.

13 The two forces in the ELBO

The ELBO can be rewritten as:

$$\text{ELBO}(q) = \underbrace{\mathbb{E}_q[\log p(x | \theta)]}_{\text{expected log-likelihood}} - \underbrace{\text{KL}(q(\theta) \| p(\theta))}_{\text{divergence from prior}} \quad (13.1)$$

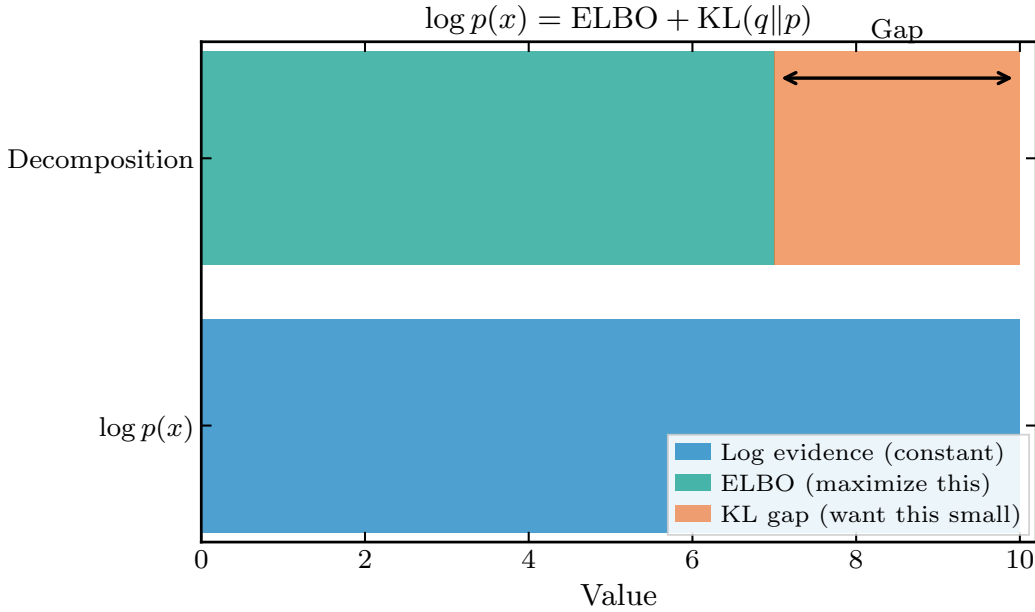


Figure 9. The two terms in the ELBO. The expected log-likelihood rewards q for placing mass where the data are well-explained. The KL from prior penalizes deviation from prior beliefs.

Two opposing forces: the first term rewards q for concentrating on parameters that explain the data well; the second penalizes q for deviating from the prior. The optimal q balances these pressures.

14 Optimizing the ELBO

The variational distribution $q(\theta; \phi)$ is parameterized by ϕ (e.g., means and variances for a Gaussian). We want $\nabla_{\phi} \text{ELBO}$ so we can do gradient ascent.

The challenge: the ELBO involves an expectation over q , which depends on ϕ . How do we differentiate through an expectation whose distribution depends on the parameters?

The reparameterization trick. Write samples as a deterministic transformation of fixed noise. For a Gaussian $q(\theta; \mu, \sigma) = \mathcal{N}(\mu, \sigma^2)$:

$$\theta = \mu + \sigma \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, 1) \quad (14.1)$$

Now θ is a deterministic function of $\phi = (\mu, \sigma)$ and noise ϵ . The distribution we're averaging over no longer depends on ϕ , so we can move the gradient inside the expectation.

Stochastic gradient ascent. In practice: sample $\epsilon \sim \mathcal{N}(0, I)$, compute $\theta = \mu + \sigma \odot \epsilon$, estimate the gradient, and update ϕ . Often a single sample suffices for a noisy but unbiased gradient estimate.

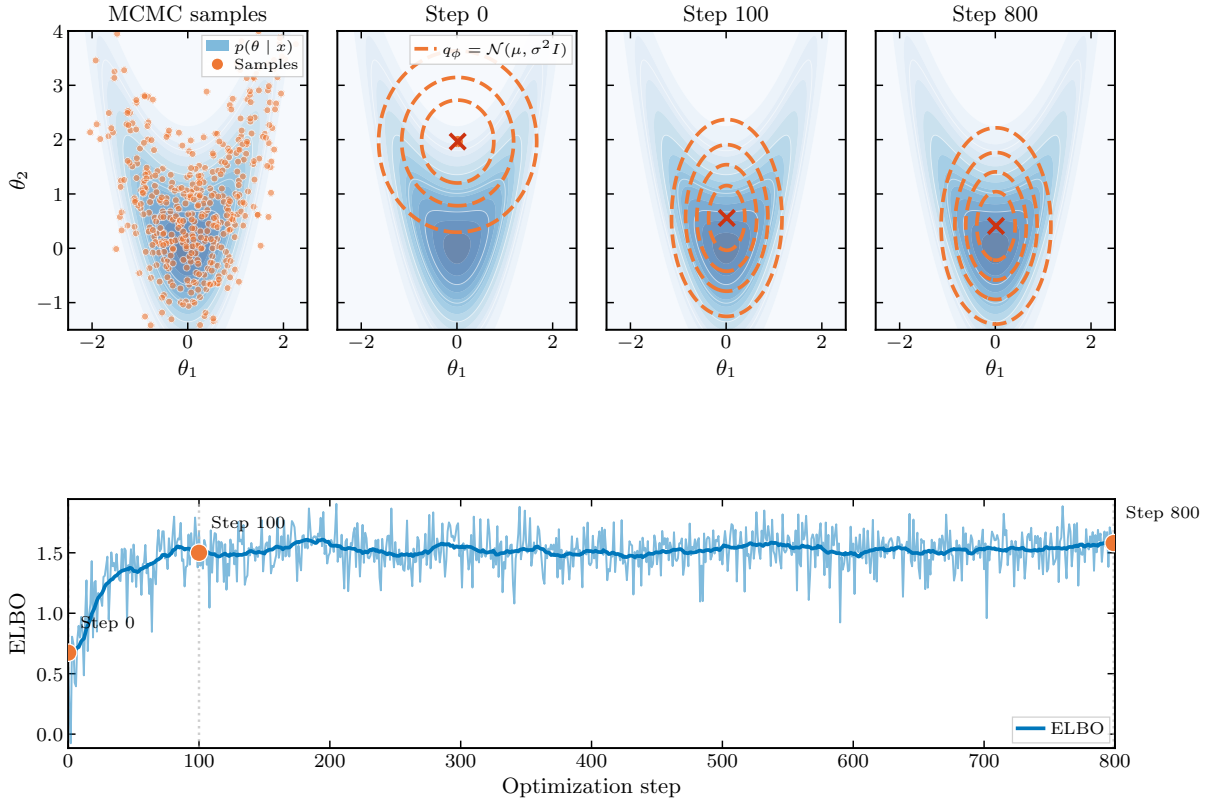


Figure 10. VI optimization. (Top left) MCMC samples from the banana posterior. (Top right) Evolution of the variational approximation during optimization. (Bottom) The ELBO increases during optimization.

15 Variational families

The choice of $\mathcal{Q}$ controls the expressiveness–tractability tradeoff. The simplest choice is mean-field Gaussians: $q(\theta) = \prod_j \mathcal{N}(\theta_j; \mu_j, \sigma_j^2)$, where each parameter is independent. This is fast but cannot capture correlations.

More expressive families—full-covariance Gaussians, normalizing flows—become important when VI is used as a building block for generative models.

16 Mode-seeking behavior

$\text{KL}(q||p)$ penalizes q for placing mass where p is small. The consequence: q is *mode-seeking*. It concentrates on one mode of the posterior and ignores others.

Because VI is mode-seeking, variational approximations typically underestimate posterior variance. If accurate uncertainty quantification is critical, MCMC may be the better choice.

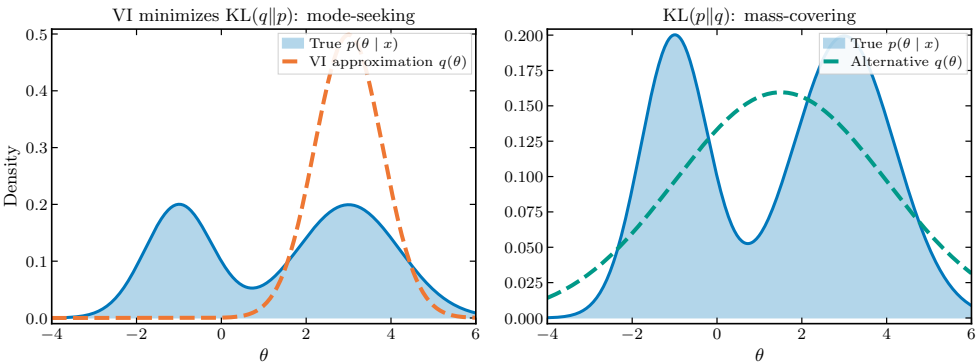


Figure 11. Mode-seeking behavior. When the posterior is multimodal, VI concentrates on a single mode rather than covering all modes.

17 Worked example: banana posterior

Consider a “banana” posterior—curved and non-Gaussian:

$$p(\theta_1, \theta_2 \mid x) \propto \exp \left(-\frac{1}{2}\theta_1^2 - \frac{1}{2}(\theta_2 - \theta_1^2)^2 \right) \tag{17.1}$$

This posterior is curved: θ_2 tends to follow θ_1^2 .

Approximate it with a mean-field Gaussian: $q(\theta) = \mathcal{N}(\theta_1; \mu_1, \sigma_1^2) \cdot \mathcal{N}(\theta_2; \mu_2, \sigma_2^2)$.

What VI gets right and wrong. The optimized q^* will:

- Correctly locate the high-probability region
- **Miss the correlation:** the mean-field assumption forces independence, so the curved banana becomes an axis-aligned ellipse
- **Underestimate uncertainty:** to avoid placing mass in the low-probability “wings,” q shrinks

A full-covariance Gaussian captures linear correlations but still misses the curvature. Capturing the banana shape requires a more flexible family (normalizing flows) or MCMC.

18 MCMC vs. VI

	MCMC	VI
Output	Samples	Approximate distribution
Accuracy	Exact (asymptotically)	Limited by family $\mathcal{Q}$
Speed	Can be slow	Fast
Multimodality	Can explore all modes	Mode-seeking
Uncertainty	Well-calibrated	Often underestimates